

Reshaping data

Warmup activity

Work on the warmup activity (on the course page), then we will discuss as a class

Warmup

```
# A tibble: 260 × 38
  country `1975` `1976` `1977` `1978` `1979` `1980`
`1983` `1984`
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl> <dbl>
1 Afghan...    NA    NA    NA    NA    4.99    NA
NA    NA
2 Albania      NA    NA    NA    NA    NA      NA
NA    NA
3 Algeria      NA    NA    NA    NA    NA      NA
NA    NA
4 Andorra      NA    NA    NA    NA    NA      NA
NA    NA
```

Question: What does a row in this data represent?

Warmup

```
# A tibble: 260 × 38
  country `1975` `1976` `1977` `1978` `1979` `1980`
`1983` `1984`
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl> <dbl>
1 Afghan...      NA      NA      NA      NA      4.99      NA
NA      NA
2 Albania        NA      NA      NA      NA      NA      NA
NA      NA
3 Algeria        NA      NA      NA      NA      NA      NA
NA      NA
4 Andorra        NA      NA      NA      NA      NA      NA
NA      NA
```

Question: What does a row in this data represent?

Each row is one country

Warmup

```
# A tibble: 260 × 38
  country `1975` `1976` `1977` `1978` `1979` `1980`
`1983` `1984`
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl> <dbl>
1 Afghan...      NA      NA      NA      NA      4.99      NA
NA      NA
2 Albania        NA      NA      NA      NA      NA      NA
NA      NA
3 Algeria        NA      NA      NA      NA      NA      NA
NA      NA
4 Andorra        NA      NA      NA      NA      NA      NA
NA      NA
```

Can I use the data in this form to make a linear model?

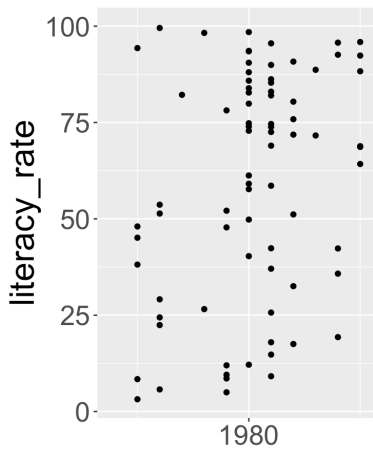
Warmup

```
# A tibble: 260 × 38
  country `1975` `1976` `1977` `1978` `1979` `1980`
`1983` `1984`
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl> <dbl>
1 Afghan...    NA    NA    NA    NA    4.99    NA
NA    NA
2 Albania     NA    NA    NA    NA    NA      NA
NA    NA
3 Algeria     NA    NA    NA    NA    NA      NA
NA    NA
4 Andorra     NA    NA    NA    NA    NA      NA
NA    NA
```

What do I want my data to look like inste

What we want

	country	year	literacy_rate
1	Afghanistan	1979	4.987460
2	Afghanistan	2011	13.000000
3	Albania	2001	98.252274
4	Albania	2008	94.681814
5	Albania	2011	95.691480
6	Algeria	1987	35.839915
7	Algeria	2002	60.075082
8	Algeria	2006	63.918785
9	Angola	2001	54.194488
10	Angola	2011	58.608460
11	Anguilla	1984	95.714930
12	Antigua and Barbuda	2001	99.420000
13	Antigua and Barbuda	2011	99.420000
14	Argentina	1980	93.580894
15	Argentina	1991	96.041358
16	Argentina	2001	97.193411
17	...	...	...



```
1 litF_long |>
2   ggplot(aes
3
4     geom_point
```

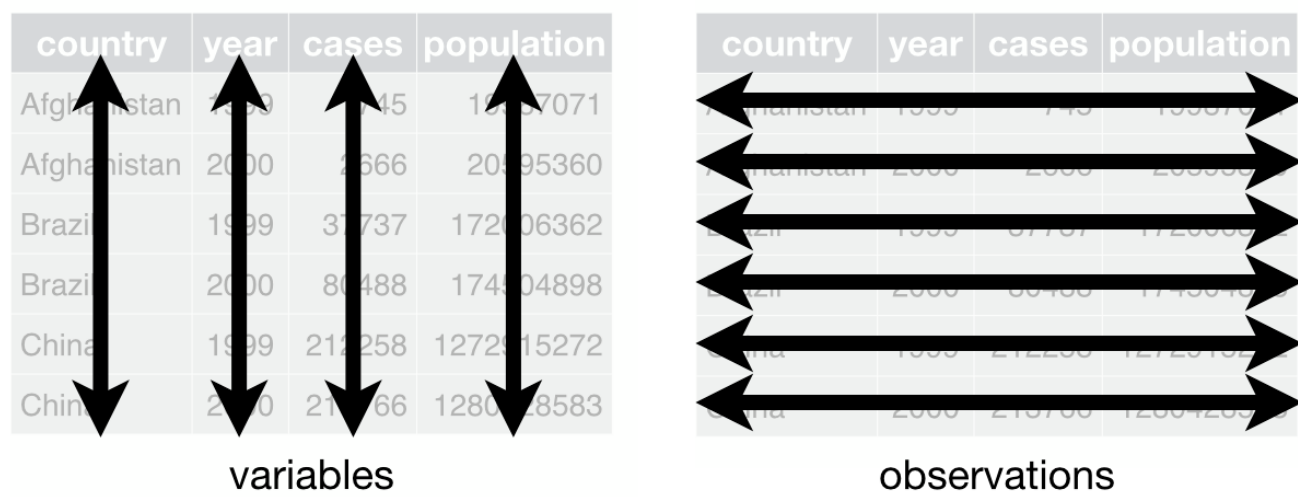
What we want

	country	year	literacy_rate
1	Afghanistan	1979	4.987460
2	Afghanistan	2011	13.000000
3	Albania	2001	98.252274
4	Albania	2008	94.681814
5	Albania	2011	95.691480
6	Algeria	1987	35.839915
7	Algeria	2002	60.075082
8	Algeria	2006	63.918785
9	Angola	2001	54.194488
10	Angola	2011	58.608460
11	Anguilla	1984	95.714930
12	Antigua and Barbuda	2001	99.420000
13	Antigua and Barbuda	2011	99.420000
14	Argentina	1980	93.580894
15	Argentina	1991	96.041358
16	Argentina	2001	97.193411
17	...	...	...

(Intercept)
-1323.2098674

```
1 lm(literacy_
```

Tidy data



(Image from *R for Data Science*)

Tidy data

Literacy data in tidy form:

```
# A tibble: 571 × 3
  country      year literacy_rate
  <chr>        <chr>      <dbl>
1 Afghanistan 1979         4.99
2 Afghanistan 2011         13
3 Albania     2001        98.3
4 Albania     2008        94.7
5 Albania     2011        95.7
6 Algeria     1987        35.8
7 Algeria     2002        60.1
8 Algeria     2006        63.9
9 Angola      2001        54.2
10 Angola     2011        58.6
```

Another example

Data on three patients (A, B, C), with two measurements (bp1 and bp2) per patient.

	id	bp1	bp2
1	A	100	120
2	B	140	115
3	C	120	125

How might we want to reshape this data?

Another example

Original data:

	id	bp1	bp2
1	A	100	120
2	B	140	115
3	C	120	125

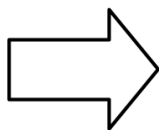
Reshaped data:

	id	measurement	value
1	A	bp1	100
2	A	bp2	120
3	B	bp1	140
4	B	bp2	115
5	C	bp1	120
6	C	bp2	125

Question: how do we do this reshaping in

Reshaping data: `pivot_longer`

id	bp1	bp2
A	100	120
B	140	115
C	120	125



id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

```
1 df |>
2   pivot_longer(
3     cols = bp1:bp2,
4     names_to = "measurement",
5     values_to = "value"
6   )
```

(Image and example from *R for Data Science*)

pivot_longer

```
# A tibble: 260 × 38
  country `1975` `1976` `1977` `1978` `1979` `1980`
`1983` `1984`
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl> <dbl>
1 Afghan...      NA      NA      NA      NA      4.99      NA
NA      NA
2 Albania      NA      NA      NA      NA      NA      NA
NA      NA
3 Algeria      NA      NA      NA      NA      NA      NA
NA      NA
4 Andorra      NA      NA      NA      NA      NA      NA
NA      NA
```

```
1 litF |>
2   pivot_longer(
3     cols = ...,
4     names_to = ...,
5     values_to = ...
6   )
```

pivot_longer

```
1 litF |>
2   pivot_longer(
3     cols = -country,
4     names_to = "year",
5     values_to = "literacy_rate"
6   )
```

A tibble: 9,620 × 3

	country <chr>	year <chr>	literacy_rate <dbl>
1	Afghanistan	1975	NA
2	Afghanistan	1976	NA
3	Afghanistan	1977	NA
4	Afghanistan	1978	NA
5	Afghanistan	1979	4.99
6	Afghanistan	1980	NA
7	Afghanistan	1981	NA
8	Afghanistan	1982	NA
9	Afghanistan	1983	NA
10	Afghanistan	1984	NA

pivot_longer

```
1 litF |>
2   pivot_longer(
3     cols = -country,
4     names_to = "year",
5     values_to = "literacy_rate"
6   ) |>
7   drop_na()
```

A tibble: 571 × 3

	country <chr>	year <chr>	literacy_rate <dbl>
1	Afghanistan	1979	4.99
2	Afghanistan	2011	13
3	Albania	2001	98.3
4	Albania	2008	94.7
5	Albania	2011	95.7
6	Algeria	1987	35.8
7	Algeria	2002	60.1
8	Algeria	2006	63.9
9	Angola	2001	54.2
10	Angola	2011	58.6

pivot_longer

```
1 litF |>
2   pivot_longer(
3     cols = -country,
4     names_to = "year",
5     values_to = "literacy_rate",
6     values_drop_na = T
7   )
```

A tibble: 571 × 3

	country <chr>	year <chr>	literacy_rate <dbl>
1	Afghanistan	1979	4.99
2	Afghanistan	2011	13
3	Albania	2001	98.3
4	Albania	2008	94.7
5	Albania	2011	95.7
6	Algeria	1987	35.8
7	Algeria	2002	60.1
8	Algeria	2006	63.9
9	Angola	2001	54.2
10	Angola	2011	58.6

Example 2

Now consider the following table:

1 ex_df					
	id	x_1	x_2	y_1	y_2
1	1	3	5	0	2
2	2	1	8	1	7
3	3	4	9	2	9

What will the following code return?

```
1 ex_df |>
2   pivot_longer(cols = -id,
3                 names_to = "group_obs",
4                 values_to = "value")
```

Example 2

Original data:

1 ex_df

	id	x_1	x_2	y_1	y_2
1	1	3	5	0	2
2	2	1	8	1	7
3	3	4	9	2	9

Reshaped data:

1 ex_df |>
2 pivot_longer(c
3 n
4 v

A tibble: 12 × 3
id group_obs v
<dbl> <chr> <
1 1 x_1
2 1 x_2
3 1 y_1
4 1 y_2
5 2 x_1
6 2 x_2
7 2 y_1
8 2 y_2
9 3 x_1
10 3 x_2

Example 2

Original data:

1 ex_df

	id	x_1	x_2	y_1	y_2
1	1	3	5	0	2
2	2	1	8	1	7
3	3	4	9	2	9

Reshaped data:

1 ex_df |>
2 pivot_longer(cols =
3 names_t
4 names_s
5 values_

A tibble: 12 × 4

	id	group	obs	value
	<dbl>	<chr>	<chr>	<dbl>
1	1	x	1	3
2	1	x	2	5
3	1	y	1	0
4	1	y	2	2
5	2	x	1	1
6	2	x	2	8
7	2	y	1	1
8	2	y	2	7
9	3	x	1	4
10	3	x	2	9

Example 3

Consider the following example data:

	id	bp_1	bp_2	hr_1	hr_2
1	1	100	120	60	77
2	2	120	115	75	81
3	3	125	130	80	93

What if we want the data to look like this:

```
# A tibble: 12 × 4
```

	id	measurement	stage	value
	<dbl>	<chr>	<chr>	<dbl>
1	1	bp	1	100
2	1	bp	2	120
3	1	hr	1	60
4	1	hr	2	77
5	2	bp	1	120
6	2	bp	2	115
7	2	hr	1	75
8	2	hr	2	81
9	3	bp	1	125

Example 3

```
1 df_3
```

	id	bp_1	bp_2	hr_1	hr_2
1	1	100	120	60	77
2	2	120	115	75	81
3	3	125	130	80	93

```
1 df_3 |>
2   pivot_longer(cols = -id,
3                 names_to = c("measurement", "stage"),
4                 names_sep = "_",
5                 values_to = "value")
```

A tibble: 12 × 4

	id	measurement	stage	value
	<dbl>	<chr>	<chr>	<dbl>
1	1	bp	1	100
2	1	bp	2	120
3	1	hr	1	60
4	1	hr	2	77
5	2	bp	1	120
6	2	bp	2	115

7	2 hr	1	75
8	2 hr	2	81
9	3 bp	1	125
10	3 hr	2	130

Example 3

```
1 df_3 |>
2   pivot_longer(cols = -id,
3                 names_to = c("measurement", "site"),
4                 names_sep = "_",
5                 values_to = "value")
```

Step 1: Pivot

A tibble: 6 × 3

	id	measurement	value
	<dbl>	<chr>	<dbl>
1	1	bp_1	100
2	1	bp_2	120
3	1	hr_1	60
4	1	hr_2	77
5	2	bp_1	120
6	2	bp_2	115

Step 2: Se

A tibble: 6 × 3

	id	measu
	<dbl>	<chr>
1	1	bp
2	1	bp
3	1	hr
4	1	hr
5	2	bp
6	2	bp

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as a PDF on Canvas (one per group, list all your names)

For next time, read:

- Chapter 5 in *R for Data Science* (2nd edition)